

>_ SESSION 02 • WEB DEVELOPMENT

HTML Deep Dive

Lists, tables, forms, links, media, and the accessibility basics that make it all usable.

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Lists & tables

2

Forms

3

Links & media

4

Accessibility

>_ Vasken Sarkis • HTML Deep Dive

Today's session

- 01 Lists & tables — structuring grouped and tabular content
- 02 Forms & validation attributes
- 03 Links, images & embedding media
- 04 Accessibility basics — alt text, labels, ARIA landmarks



ALSO TODAY

Lab: build a contact form and an image gallery using only semantic HTML.

01

Lists & Tables

Two ways to structure content that has multiple parts — grouped items, and rows of data.

Three kinds of lists

unordered



```
<ul>
  <li>HTML</li>
  <li>CSS</li>
</ul>
```

Order doesn't matter — bullet points

ordered



```
<ol>
  <li>Mix</li>
  <li>Bake</li>
</ol>
```

Sequence matters — numbered steps

description



```
<dl>
  <dt>DOM</dt>
  <dd>Document tree</dd>
</dl>
```

Pairs a term with its definition

Tables hold tabular data

```
schedule.html

1 <table>
2   <thead>
3     <tr>
4       <th>Day</th><th>Topic</th>
5     </tr>
6   </thead>
7   <tbody>
8     <tr><td>Mon</td><td>HTML</td></tr>
9     <tr><td>Wed</td><td>CSS</td></tr>
10  </tbody>
11 </table>
```



NOTE

<th> marks a header cell; <td> marks a regular data cell.



GOTCHA

Tables are for data with rows and columns — never reach for a table to lay out a whole page.

Day	Topic
Mon	HTML
Wed	CSS
...	

02

Forms

How a page collects input — and how the browser can catch mistakes before they're ever submitted.

Anatomy of a form

contact.html

```
1 <form action="/submit" method="post">
2   <label for="email">Email</label>
3   <input id="email" name="email" type="email">
4   <label for="msg">Message</label>
5   <textarea id="msg" name="msg"></textarea>
6   <button type="submit">Send</button>
7 </form>
```



KEY IDEA

Every `<input>` needs a `<label>`, connected by matching `for` / `id`.



TIP

Clicking the label text focuses the input — a free usability win.

Validation attributes catch mistakes early

contact.html

```
1 <input type="email" required>
2
3 <input type="tel"
4   pattern="[0-9]{10}">
5
6 <input type="number"
7   min="1" max="5">
```

required

type

pattern

min / max



NOTE

The right type (email, tel, number) gets free validation and the right on-screen keyboard on mobile.



GOTCHA

Built-in validation is a first line of defense, not the only one — always re-check submitted data on the server too.

03

Links, Images & Media

Connecting pages together, and bringing in the images, video, and audio that make them richer.

Links and images

gallery.html

```
1 <a href="https://github.com">
2   GitHub
3 </a>
4
5 <a href="about.html">About</a>
6
7 
```



NOTE

An absolute URL points anywhere on the web; a relative path like `about.html` points within your own site.



KEY IDEA

`alt` text isn't optional — it's read aloud by screen readers and shown if the image fails to load.

Embedding media

video



```
<video src="clip.mp4"  
controls></video>
```

Native player with play, pause,
volume

audio



```
<audio src="song.mp3"  
controls></audio>
```

Same idea, audio-only controls

iframe



```
<iframe  
  src="https://maps...">  
</iframe>
```

Embeds an entire other page, like a map

04

Accessibility Basics

Alt text, labels, and landmarks — small habits that make a page usable for everyone.

Three habits that go a long way

- 01 Every `img` gets an `alt` describing what it shows, or an empty `alt=""` if it's purely decorative.
- 02 Every form field gets a real `<label>`, not just placeholder text.
- 03 Semantic landmarks — `header`, `nav`, `main` — let screen readers jump straight to a section.



KEY IDEA

Accessible HTML is usually just correct HTML — the right tag for the right job, most of the way there for free.

ARIA fills gaps, it doesn't replace HTML

page.html

```
1 <nav aria-label="Main">...</nav>
2
3 <div role="search">
4   <input type="search">
5 </div>
6
7 <!-- semantic <nav> already has a
8     landmark role – this role='search'
9     div doesn't have a native tag yet -->
```



TIP

Rule one of ARIA: if a native tag already does the job — nav, button — use it before reaching for a role attribute.



NOTE

Use `aria-label` to disambiguate when a page has more than one of the same landmark.

Build a contact form & a gallery

Two small pages, semantic HTML only — no styling yet.

1. Contact form: name, email, message, with labels and **required**
2. Image gallery: a grid of `<img>` tags, every one with real alt text
3. Wrap each page in header/nav/main/footer

forms

alt text

no styling yet

Use the browser's built-in validation — try submitting an empty form.

Extend the About Me page

HTML only

no CSS yet

single page

- Add a bio section with at least one heading and two paragraphs
- Add a real photo with meaningful `alt` text
- Add a contact form: name, email, message — all labeled, **required** where it matters
- Commit and push each meaningful change



DUE

Start of Session 3. Next class we style all of this with CSS — colors, type, and spacing.